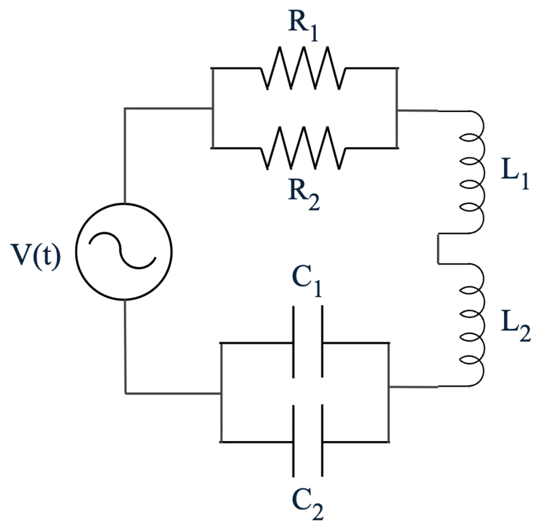
Department of Physics, FNSPE, CTU in Prague

02YELMA - Electricity and Magnetism

Final Exam

15.6.2026 / 14:00 - 16:30

Q1: Resistors, inductors, and capacitors are connected with an alternating voltage source $V(t) = V_0 \sin(\omega t)$ as shown in the figure. Assume that the circuit is ideal and there is no mutual inductance between the inductors.



(a) Write down the differential equation governing the current $I(t)$ in the circuit.

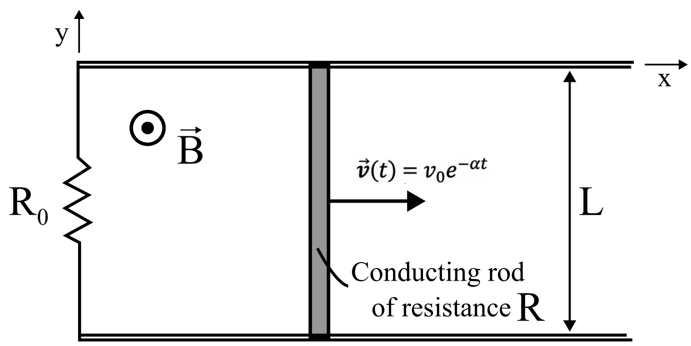
(b) To obtain the greatest possible current amplitude, what should be the value of ω?

(c) Given that the steady-state current is $I(t) = I_0 \sin(\omega t + \phi)$, find the average energy dissipation in the circuit at the resonance frequency per unit time?

Q2: A long coaxial capacitor consists of two concentric cylindrical conductors with radii a and b ($a < b$) and vacuum spacing. It is driven by a time-dependent voltage $V(t) = V_0 \cos(\omega t)$ applied between the inner and outer conductors. Assume cylindrical symmetry and neglect edge effects. Determine the electric field $\vec{E}(r,t)$ in the regions $a < r < b$ and $r > b$.

Q3: A parallel-plate capacitor is at rest in frame S', producing a uniform electric field $\vec{E}' = E_0 \hat{y}'$. Frame S' moves with constant velocity $\vec{v} = v\hat{x}$ relative to laboratory frame S. Determine the electric field $\vec{E}$ and magnetic field $\vec{B}$ in frame S interpret the result in terms of the motion of the charges in the capacitor. *Given:* $E_\parallel = E'_\parallel$, $E_\perp = \gamma(E'_\perp - \vec{v} \times \vec{B}')$, $B_\parallel = B'_\parallel$, and $B_\perp = \gamma(B'_\perp + \frac{1}{c^2}\vec{v} \times \vec{E}')$, where $\gamma = 1/\sqrt{1 - v^2/c^2}$ is the Lorentz factor and directions $\parallel$ and $\perp$ are defined with respect to the velocity $\vec{v}$.

Q4: A conducting rod of length L and resistance R slides without friction on two parallel conducting rails that are connected by a resistor R_0, forming a closed rectangular circuit. The entire setup lies in the xy-plane. A uniform magnetic field $\vec{B} = B_0 \hat{z}$ is directed perpendicular to the plane of the circuit. Initially, the rod moves to the right with velocity $v(t) = v_0 e^{-\alpha t}$ where v_0 and α are positive constants. Neglect the resistance of the rails.



(a) Determine the motional electromotive force (EMF) induced in the circuit and the induced current $I(t)$ (including its direction) as a function of time.

(b) Calculate the magnetic force acting on the rod as a function of time. Verify that its direction is consistent with Lenz's law.

(c) Suppose the prescribed velocity is maintained by an external agent acting on the rod (of mass m), determine the electrical power P_{elec} dissipated in the circuit and the mechanical power P_{ext} supplied by the agent verify the energy conservation and comment on the sign of P_{ext}.